

SUMMARY

This topic covers advanced Number Systems concepts for General Aptitude, with particular emphasis on Unit Digit Value (UDV) and Divisibility Rules. It focuses on using Power Cycles (cyclicity) to determine the last digit of large exponential expressions and structured divisibility techniques to test large integers efficiently. The approach emphasizes repetition patterns of unit digits, normalization of negative results, factorial-based simplifications, and decomposition of composite divisors into co-prime components for precise testing.

This video synthesizes key instructional insights regarding Number Systems, specifically focusing on Unit Digit Value (UDV) and Divisibility Rules. The core methodology relies on the concept of Power Cycles to determine the last digit of large exponential numbers and the application of structured divisibility tests for complex integers. Major takeaways include the four-step cyclic repetition of most unit digits, normalization of negative unit digits by adding multiples of ten, simplification of factorial exponents due to inherent divisibility properties, and testing composite divisors such as 72 by decomposing them into co-prime factors 8 and 9.

1. UNIT DIGIT VALUE (UDV) AND POWER CYCLES

The Unit Digit Value of any exponential expression depends solely on the last digit of the base and the cyclic behavior of its powers.

1.1 THE CONCEPT OF CYCLICITY

Cyclicity, also known as the Power Cycle (P_c), defines how frequently the unit digit repeats as powers increase. To determine the UDV:

1. Identify the last digit of the base.
2. Determine its power cycle length.
3. Divide the exponent by the power cycle.
4. Use the remainder to identify the correct unit digit.

Power Cycle Classification:

Digits: 0, 1, 5, 6

Power Cycle: 1

Characteristic: Unit digit remains constant for any exponent.

Digits: 4, 9

Power Cycle: 2

Characteristic: Alternates between two digits depending on whether the exponent is odd or even.

Digits: 2, 3, 7, 8

Power Cycle: 4

Characteristic: Follows a four-step repetition pattern.

1.2 SPECIFIC DIGIT RULES

For 2 and 8:

Power cycle is 4.

If the exponent is a multiple of 4, $UDV = 6$.

For 3 and 7:

Power cycle is 4.

If exponent is a multiple of 4, $UDV = 1$.

For 4:

4 raised to an odd power $\rightarrow UDV = 4$.

4 raised to an even power $\rightarrow UDV = 6$.

For 9:

9 raised to an odd power $\rightarrow UDV = 9$.

9 raised to an even power $\rightarrow UDV = 1$.

The "Six" Property:

Multiplying 6 by any even number produces a product whose unit digit equals that even number.

Example:

$$6 \times 4 = 24$$

$$6 \times 8 = 48$$

2. ADVANCED UDV CALCULATIONS

Complex UDV problems may involve subtraction, factorial exponents, or trailing zero components.

2.1 NEGATIVE UNIT DIGITS

If subtraction results in a negative unit digit, normalize it by adding the next multiple of 10.

Example:

If result = -2

Add 10 $\rightarrow$ Final UDV = 8.

This ensures the unit digit remains between 0 and 9.

2.2 FACTORIALS IN POWERS

Factorials simplify UDV evaluation due to built-in divisibility:

Divisibility by 4:

Any factorial $n!$ where $n \geq 4$ is divisible by 4.

This guarantees completion of the power cycle for digits with cycle length 4.

Evenness:

Any factorial $n!$ where $n \geq 2$ is always even.

Thus, factorial exponents frequently reduce to full-cycle cases.

2.3 RIGHT-MOST NON-ZERO DIGIT

To find the right-most non-zero digit of 30^{2720} :

Step 1:

Rewrite as $(3 \times 10)^{2720}$

$= 3^{2720} \times 10^{2720}$.

Step 2:

Ignore 10^{2720} since it contributes only trailing zeros.

Step 3:

Evaluate UDV of 3^{2720} .

Since 2720 is divisible by 4, the UDV is 1.

3. DIVISIBILITY TESTS

Divisibility rules enable quick factor testing without long division.

3.1 STANDARD DIVISIBILITY RULES

Divisibility by 3:

Sum of digits must be divisible by 3.

Divisibility by 4:

Last two digits must be divisible by 4.

Divisibility by 8:

Last three digits must be divisible by 8.

Divisibility by 9:

Sum of digits must be divisible by 9.

Divisibility by 11:

Difference between the sum of digits at odd positions and even positions must be 0 or a multiple of 11.

3.2 COMPOSITE DIVISORS (THE 72 RULE)

Since $72 = 8 \times 9$ and 8 and 9 are co-prime:

A number is divisible by 72 only if it satisfies divisibility by both 8 and 9.

Step 1:

Check divisibility by 8 using the last three digits.

Step 2:

Check divisibility by 9 using the sum of all digits (including variables).

Both conditions must hold simultaneously.

4. INSTRUCTIONAL CONTEXT AND LOGISTICS

This instructional framework is delivered by Gurupal Chawla (GC Sir), an educator specializing in Mathematics and Aptitude.